21. Which one of the following is true about metallic character when we move from left to right in a period and top to bottom in a group?

a) Decreases in a period and increases along the group

b) Increases in a period and decreases in a group

c) Increases both in the period and the group

d) Decreases both in the period and in the group

22. How does electron affinity change when we move from left to right in a period in the periodic table?

a) Generally increases b) Generally decreases

c) Remains unchanged d) First increases and then decreases

23. Which of the following pairs of elements exhibit diagonal relationship?

a) Be and Mg b) Li and Be c) Be and B d) Be and Al

II. Write brief answer to the following questions

24. Define modern periodic law.

25. What are isoelectronic ions? Give examples.

26. What is effective nuclear charge ?

27. Is the definition given below for ionisation enthalpy is correct?

"Ionisation enthalpy is defined as the energy required to remove the most loosely bound electron from the valence shell of an atom"

28. Magnesium loses electrons successively to form Mg^{+}, Mg^{2+} and Mg^{3+} ions. Which step will have the highest ionisation energy and why?

29. Define electronegativity.

30. How would you explain the fact that the second ionisation potential is always higher than first ionisation potential?

31. Energy of an electron in the ground state of the hydrogen atom is -2.18×10^{-18} J. Calculate the ionisation enthalpy of atomic hydrogen in terms of kJ mol^{-1}.

32. The electronic configuration of atom is one of the important factor which affects the value of ionisation potential and electron gain enthalpy. Explain

33. In what period and group will an element with Z = 118 will be present?

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